PEARSON EDEXCEL INTERNATIONAL A LEVEL · CHEMISTRY · WCH12

07

Intermolecular Forces

London forces, dipoles and hydrogen bonds — the forces between molecules.

London forces and electron density; permanent dipole–dipole interactions; hydrogen bonding in H_2O, NH_3 and HF; anomalous properties of water; boiling trends of alkanes, alcohols and hydrogen halides; solvents and solubility.

UNIT	SECTIONS	STUDY TIME	FLASHCARDS
2 · WCH12	7	~3.5 h	14

COVERS **7 SPECIFICATION POINTS** — STATEMENTS QUOTED VERBATIM FROM THE SPECIFICATION

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7.6	Solvents and Solubility	SPEC 7.6	35 min

HOW TO USE THIS DOCUMENT

Every section opens with the specification statement, worded exactly as Edexcel words it — that is what the examiner marks against. Definitions, equations and exam alerts are boxed; worked examples walk the full method step by step. Read a section, shut the page and recall its bullets from memory, then grind the topic's flashcard deck on the site.

SPEC 7.1 - 7.6 ≈ 10 MIN

Specification Map

THE SPECIFICATION SAYS

Students will be assessed on their ability to:

Every statement the specification assesses in this topic — use it as your revision checklist.

SPEC	ASSESSMENT STATEMENT
7.1	understand the nature of the following intermolecular forces: i London forces (instantaneous dipole–induced dipole) ii permanent dipole–permanent dipole interactions iii hydrogen bonds
7.2	understand the interactions in molecules, such as H_2O, liquid NH_3 and liquid HF, which give rise to hydrogen bonding
7.3	understand the following anomalous properties of water resulting from hydrogen bonding: i its high melting and boiling temperature when compared with similar molecules ii the density of ice compared to that of water
7.4	be able to predict the presence of hydrogen bonding in molecules analogous to those mentioned in 7.2
7.5	understand, in terms of intermolecular forces, physical properties shown by substances, including: i the trends in boiling temperatures of alkanes with increasing chain length ii the effect of branching in the carbon chain on the boiling temperatures of alkanes iii the relatively low volatility (higher boiling temperatures) of alcohols compared to alkanes with a similar number of electrons iv the trends in boiling temperatures of the hydrogen halides HF to HI
7.6	understand factors that influence the choice of solvents, including: i water, to dissolve some ionic compounds, in terms of the hydration of the ions ii water, to dissolve simple alcohols, in terms of hydrogen bonding iii water, as a poor solvent for compounds (to include polar molecules such as halogenoalkane), in terms of inability to form hydrogen bonds iv non-aqueous solvents, for compounds that have similar intermolecular forces to those in the solvent

This topic builds directly on the foundations of Topic 3: **electronegativity** (3.13), **bond polarity** (3.14) and the shapes of **polar molecules** (3.15). Here the polarity of bonds is used to explain the attractions **between** whole molecules — and those attractions turn out to decide how substances boil, melt, mix and dissolve.

SPEC 7.1 ≈ 40 MIN

The Nature of Intermolecular Forces

THE SPECIFICATION SAYS

understand the nature of the following intermolecular forces: i London forces (instantaneous dipole–induced dipole) ii permanent dipole–permanent dipole interactions iii hydrogen bonds

London forces, permanent dipole–dipole interactions and hydrogen bonds — the three non-bonded interactions.

DEFINITION

Intermolecular forces

The weak **electrostatic attractions between molecules** — described as **non-bonded interactions** because they occur *between* molecules, not inside them. They are considerably weaker than typical covalent and polar covalent bonds.

- Intermolecular forces act **in addition to** the covalent bonds inside each molecule — they do not replace them.
- Because they are weak, they are overcome easily: they decide **physical properties** such as melting temperature, boiling temperature, volatility and solubility.

The three types of intermolecular force

- **Instantaneous dipole–induced dipole interactions** — also called **London forces**, temporary dipole forces or dispersion forces.
- **Permanent dipole–permanent dipole interactions** — between molecules with a permanent separation of charge.
- **Hydrogen bonds** — the strongest of the three; a special case of dipole–dipole attraction involving hydrogen.

London forces

DEFINITION

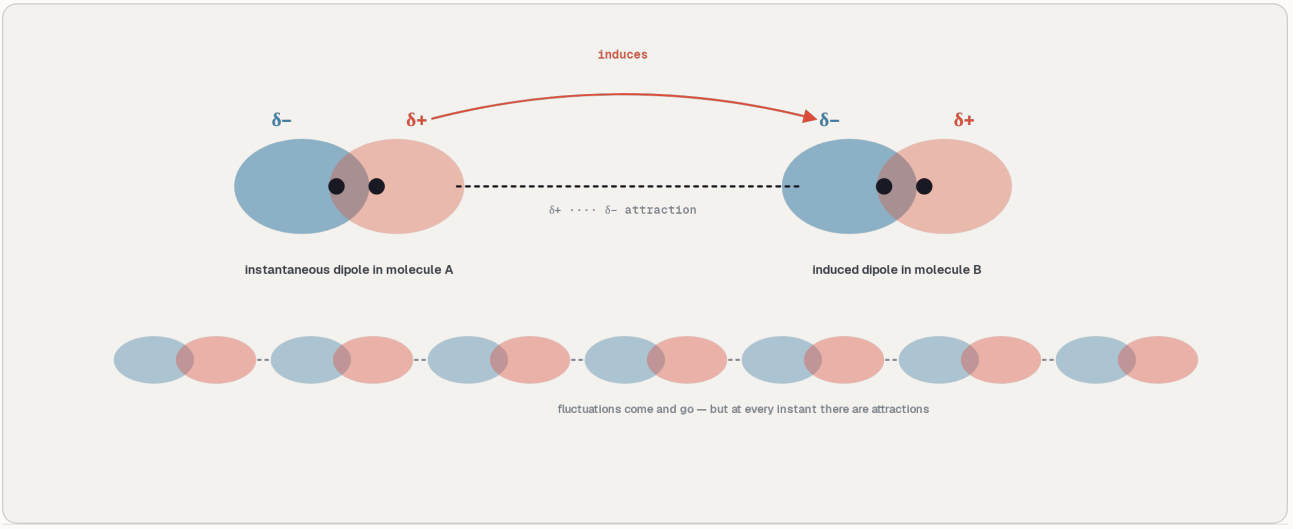
London forces

Instantaneous dipole–induced dipole attractions. They occur between **all molecular substances and noble gases** — they do not occur in ionic substances.

- In any molecule the electrons are moving **constantly and randomly**. As this happens the electron density can fluctuate, so parts of the molecule become more or less negative — small **temporary (transient) dipoles** form.
- These temporary dipoles can cause dipoles to form in **neighbouring molecules** — these are called **induced dipoles**. The induced dipole is always the **opposite sign** to the original one on the facing sides.
- The two dipoles between the molecules are arranged so that they interact favourably with one another — this favourable interaction is the **London force of attraction**.

- London forces exist between **all** simple covalent molecules and the separate atoms in noble gases — no molecule can escape them.

SPEC 7.1 · LONDON FORCES **Instantaneous dipole – induced dipole attraction** all molecules · noble gases



ORIGIN Electrons move constantly and randomly; the electron density fluctuates and a temporary (transient) dipole forms. It induces an opposite dipole in its neighbour — the facing δ+ / δ− ends attract. No molecule can escape London forces; ionic substances have none.

FIG – *The fluctuating electron density in molecule A creates an instantaneous dipole, which induces an opposite dipole in its neighbour B — the facing δ+ and δ− ends attract.*

Feature 1 — the size of the electron cloud

- The **more electrons** there are in a molecule, the greater the fluctuations in electron density and the larger the instantaneous and induced dipoles created.
- This makes the London forces stronger between the molecules, so **more energy is needed to break them** — boiling temperatures will be greater.
- This is demonstrated by the boiling temperatures of the **noble gases**: they are monatomic (single atoms), so London forces are the **only** force of attraction between the particles.

GAS	HELIUM	NEON	ARGON	KRYPTON	XENON	RADON
Boiling temperature / K	4.3	27.1	87.4	121	165	211

- Down Group 18 the atoms get bigger and hold **more electrons**, so the London forces between the atoms increase — and the boiling temperature rises.

KEY POINT

The first rule of London forces

As the **size of the electron cloud** increases, the **strength of London forces increases** — more electrons, bigger dipoles, higher boiling temperature.

Feature 2 — the area of contact between molecules

- The **shape** of the molecule also affects the size of the London forces: the more **points of contact** there are between molecules, the greater the overall London force.
- A long, straight chain lies alongside its neighbours over its whole length; a compact, branched molecule touches its neighbours at fewer points.
- Pentane and 2,2-dimethylpropane share the same molecular formula C_5H_{12} — the same electrons, but very different shapes — so the difference in their boiling temperatures is caused almost entirely by the **contact area**.

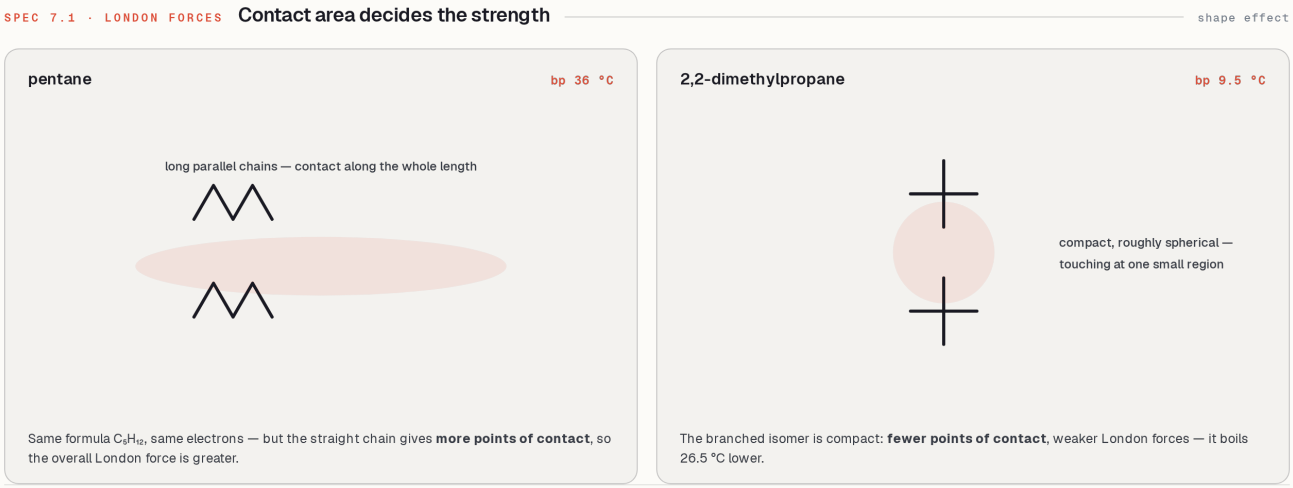


FIG — *More points of contact → stronger London forces → higher boiling temperature: pentane boils at 36 °C, its compact isomer 2,2-dimethylpropane at only 9.5 °C.*

Permanent dipole–dipole interactions

DEFINITION

Permanent dipole

A **permanent separation of charge** across a molecule, produced by polar bonds arranged unsymmetrically — one end of the molecule is always δ−, the other always δ+.

- Polar molecules are attracted towards one another by **permanent dipole–dipole interactions**: the negative end of one molecule attracts the positive end of another. For example, Hδ+—Clδ− molecules line up δ+→δ−.
- These interactions occur between molecules containing permanent dipoles **but which cannot form hydrogen bonds**.
- As a force *type*, permanent dipole–dipole interactions are weaker than hydrogen bonding but stronger than the weak van der Waals (London) forces.
- In a liquid, however, the **random movement** of the molecules means the dipoles are not always aligned to produce a favourable interaction — sometimes like ends meet and the pair repels.
- As a result, **averaged out**, the interaction between permanent dipoles is usually **less** than the interaction between instantaneous and induced dipoles — the **London force is usually the more significant** interaction

between molecules.

WATCH OUT

Do not dismiss London forces

It is tempting to assume the permanent dipole always dominates. In real liquids the permanent dipoles spend much of their time misaligned, so London forces — acting all the time, between all molecules — are usually the **bigger** contributor to the total attraction.

Relative strengths of intermolecular forces

- Intermolecular forces are **much weaker than intramolecular forces** (the bonds *within* molecules): covalent bond strengths are typically between **200 to 500 kJ mol^{-1}**.
- In comparison, the strength of **hydrogen bonds** is about **5 to 40 kJ mol^{-1}**.
- **Permanent dipole–dipole interactions** are weaker than hydrogen bonds, and **London forces** are still weaker.

```
hydrogen bonding  >  permanent dipole-dipole interactions  >  London forces
```

The strength order to quote — as force types

TIP

Forces affect physical properties — not chemical properties

Intermolecular forces are **not strong enough to influence chemical properties**, but they **can affect physical properties**. When a molecular substance boils, only the forces **between** molecules are overcome — the covalent bonds inside each molecule stay intact (7.5). Reacting breaks the bonds; boiling breaks the forces.

SPEC 7.2 ≈ 25 MIN

Hydrogen Bonding in Water, Ammonia and Hydrogen Fluoride

THE SPECIFICATION SAYS

understand the interactions in molecules, such as H_2O, liquid NH_3 and liquid HF, which give rise to hydrogen bonding

The strongest intermolecular force — where it comes from, and how to draw it.

DEFINITION

Hydrogen bond

The strong intermolecular attraction between a **lone pair** of electrons on a nitrogen, oxygen or fluorine atom of one molecule and a **hydrogen atom covalently bonded to N, O or F** in another molecule. It is the strongest type of intermolecular force.

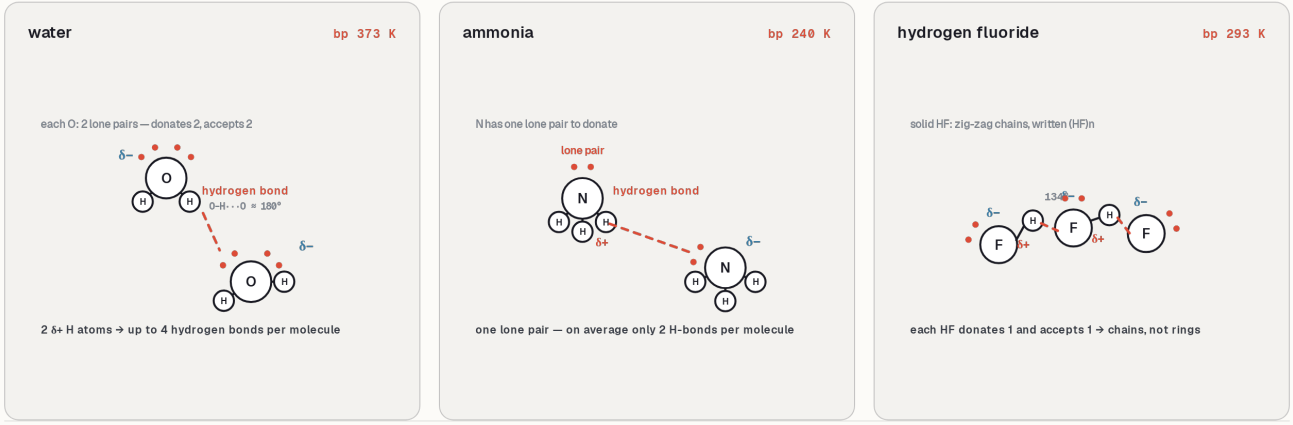
- Hydrogen bonding occurs in compounds that have a hydrogen atom covalently attached to one of the **three most electronegative atoms — nitrogen, oxygen and fluorine only**, which has a **lone pair** of electrons.

- So hydrogen bonding is found where there is an **O–H**, **N–H** or **F–H** bond — there is a large electronegativity difference between the H and the O, N or F.
- The classic examples are **water H_2O**, **ammonia NH_3** and **hydrogen fluoride HF**.
- Bonds between hydrogen and these three elements are **extremely polar**.

Why the hydrogen bond forms

- Due to the big difference in electronegativity between hydrogen and F, O or N, a substantial amount of electron density is drawn away from the hydrogen atom, leaving it with a **δ+ charge**.
- The hydrogen atom **does not have an inner shell of electrons** — as a result, its nucleus is **exposed** by the shift in electron density and is not screened.
- The **lone pair** of electrons on the F, O or N atom of another molecule is attracted to this positive hydrogen — leading to the formation of a **hydrogen bond**.
- The interaction is **not just an extreme dipole–dipole attraction**: there is some **partial bond formation** using the lone pair of electrons on the O, N or F atom.
- Hydrogen bonds are **directional** — the angle between the three atoms involved (X–H···Y) is often **180° or close to it**, which is why hydrogen-bonded structures have definite geometries.

SPEC 7.2 · HYDROGEN BONDING **Drawing the hydrogen bond — N, O and F only** δ+ H ··· lone pair



GEOMETRY The lone pair on N/O/F of one molecule is attracted to the exposed δ+ hydrogen of an O–H, N–H or F–H bond in another. The bond is directional — X–H···Y is usually close to 180°. Draw the lone pairs and the δ charges every time.

FIG – *Drawing hydrogen bonds: lone pairs shown on N, O and F, the δ+/δ− charges marked, the H-bond dashed. Water offers two lone pairs and two δ+ hydrogens; ammonia offers one lone pair; HF forms zig-zag chains (HF)n in the solid state.*

Hydrogen bonding through oxygen

- All compounds containing an **–O–H group** form intermolecular hydrogen bonds.
- In water, the hydrogen bond forms between the **oxygen atom** of one water molecule and the **hydrogen atom** of a second water molecule.
- **Alcohols** also form intermolecular hydrogen bonds through their –O–H group.

Hydrogen bonding through nitrogen

- All compounds containing an **–N–H group** can form intermolecular hydrogen bonds.
- Examples: **ammonia NH_3**, and organic compounds such as **primary amines**, which have the general formula **RNH_2**.

Hydrogen bonding through fluorine

- The **only** fluorine compound with intermolecular hydrogen bonding is **hydrogen fluoride, HF**.
- In the solid state HF molecules are held in **zig-zag chains**, written **(HF)n** — each molecule donates one hydrogen bond through its H and accepts one through a lone pair on its F.

How many hydrogen bonds per molecule?

MOLECULE	LONE PAIRS ON N/O/F	δ+ H ATOMS ON N/O/F	HYDROGEN BONDS PER MOLECULE
H_2O	2	2	up to 4
NH_3	1	3	on average 2 — one through the lone pair, one through a δ+ hydrogen
HF	3	1	2 — one donated, one accepted, giving chains

EXAM ALERT

The HF vs H_2O paradox

Hydrogen bond **strength** for HF is greater than for H_2O, yet HF boils lower (293 K vs 373 K). Why? **1)** HF forms two hydrogen bonds per molecule, whereas water can form up to **four** — hydrogen bonding in water is much more extensive. **2)** Not all the hydrogen bonds in HF are broken on vaporisation, since HF is substantially **polymerised even in the gas phase**.

- **Ammonia has the lowest boiling temperature** of the three compounds. The amount of hydrogen bonding in ammonia is limited because each nitrogen atom has only **one lone pair** of electrons — on average an ammonia molecule can form one hydrogen bond using the lone pair on nitrogen, and one hydrogen bond involving one of its δ+ hydrogen atoms.

SPEC 7.3 ≈ 30 MIN

Anomalous Properties of Water

THE SPECIFICATION SAYS

understand the following anomalous properties of water resulting from hydrogen bonding: i its high melting and boiling temperature when compared with similar molecules ii the density of ice compared to that of water

Two famous quirks of water, both explained by hydrogen bonding.

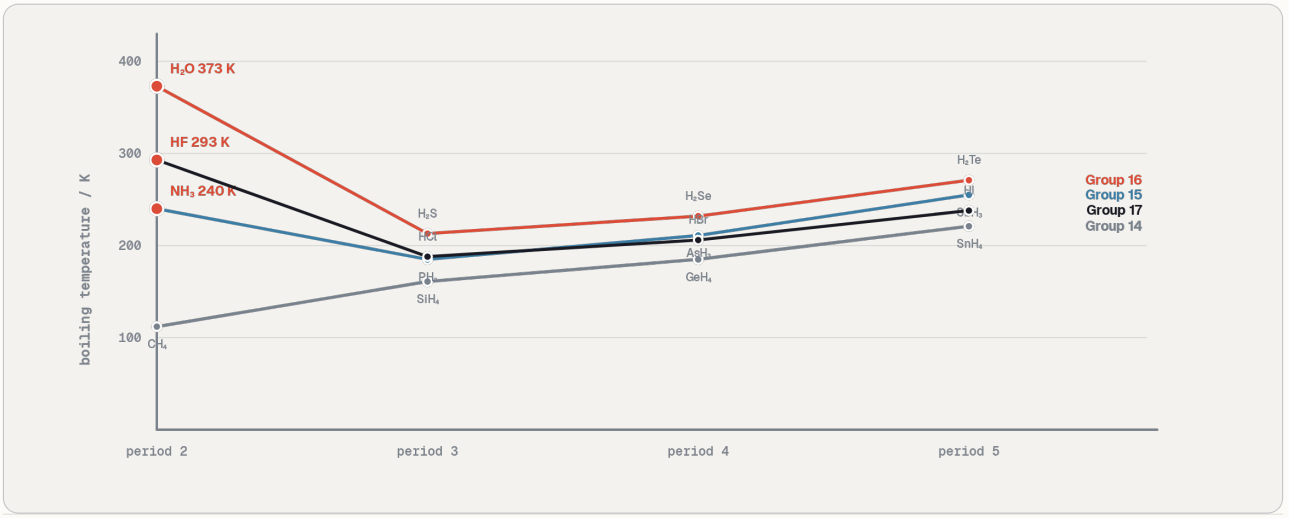
Water shows two **anomalous** (unexpected) properties, and both are consequences of hydrogen bonding:

1. It has a **relatively high melting and boiling temperature** for a molecule with so few electrons.
2. The **density of ice** at 0 °C is **less** than that of water at 0 °C.

Anomaly i — the high melting and boiling temperatures

- The hydrogen bonds between water molecules are relatively strong.
- Water therefore has an abnormally **high melting temperature (0 °C, 273 K)** and **boiling temperature (100 °C, 373 K)** at 100 kPa pressure — far higher than the hydrides of the elements around oxygen in the periodic table.

SPEC 7.3 · THE HYDRIDE GRAPH **Boiling temperatures of the hydrides — three anomalies** H_2O · NH_3 · HF



EVIDENCE Along every series the boiling temperature climbs as the electron cloud grows — but H_2O, NH_3 and HF sit far above their group trends. Their molecules are hydrogen bonded in addition to London forces, and the extra forces need more energy to break.

FIG — *The hydride boiling-point graph. Each Group 14–17 series rises steadily as the electron cloud grows — but H_2O, NH_3 and HF sit far above their trends because of hydrogen bonding. Water is the most extreme anomaly of all.*

- In **Group 14 (IV) hydrides** the boiling point increases as the electron cloud of the molecule increases and relative molecular mass increases — so the strength of the van der Waals (London) forces increases.
- The **anomalously high boiling points of H_2O, NH_3 and HF** are caused by **hydrogen bonding** between these molecules **in addition to** their London forces. The additional forces require more energy to break, and so they have higher boiling points.

TIP

Why the anomaly matters so much

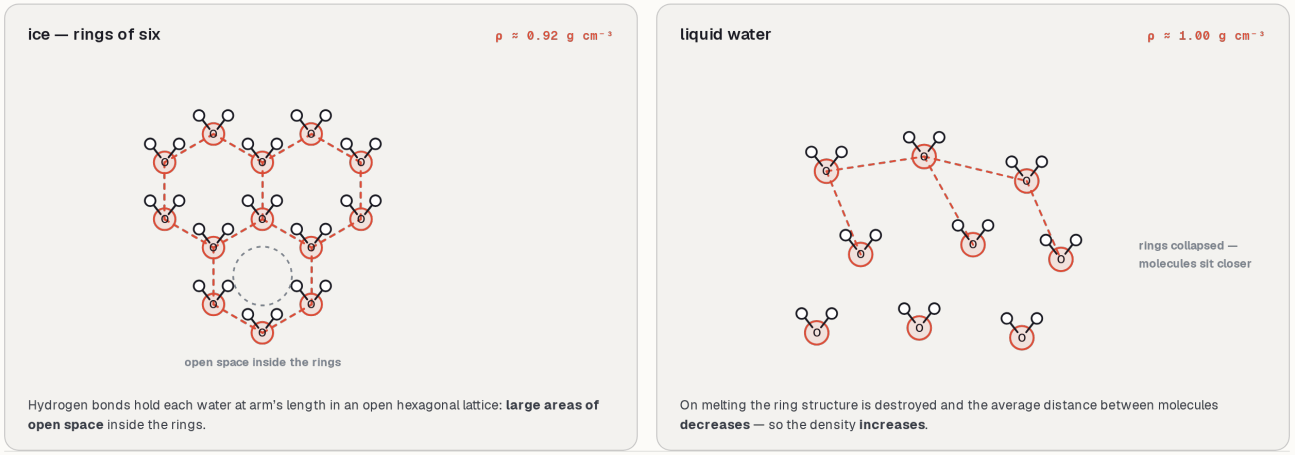
Without hydrogen bonding, water would boil at roughly −80 °C by extrapolation of the Group 16 trend (H_2S 213 K → H_2Te 271 K). There would be **no liquid water on Earth** — and no aqueous chemistry, no oceans and no life as we know it.

Anomaly ii — the density of ice

- The molecules in **ice** are arranged in **rings of six**, held together by hydrogen bonds.

- In ice there are **large areas of open space inside the rings** — the open, honeycomb lattice holds the molecules further apart than they would prefer to be.
- When ice melts, the ring structure is **destroyed** and the average distance between molecules **decreases**, causing an **increase in density**.
- So ice is **less dense** than liquid water at the same temperature — which is why ice floats on water.

SPEC 7.3 · DENSITY OF ICE **Why ice floats on water** anomaly ii



STRUCTURE Ice at 0 °C is less dense than water at 0 °C — the open hydrogen-bonded rings lock the molecules further apart. Ice floats, lakes freeze from the top down, and aquatic life survives below the ice layer.

FIG — *In ice the hydrogen-bonded rings of six lock the molecules into an open lattice. On melting the rings collapse, the molecules pack closer — and the density rises.*

KEY POINT

Why the ice anomaly matters

Because ice floats, a freezing lake **cools from the top down**: the ice layer insulates the water beneath, so a pond freezes from the surface inwards and aquatic life survives in the denser 4 °C water below.

SPEC 7.4 ≈ 15 MIN

Predicting Hydrogen Bonding

THE SPECIFICATION SAYS

be able to predict the presence of hydrogen bonding in molecules analogous to those mentioned in 7.2

A two-question test — then apply it to alcohols, amines, acids and ethers.

The specification asks you to predict hydrogen bonding in molecules **analogous to** H_2O, NH_3 and HF. Apply a two-question test to any molecule:

1. **Does the molecule contain an O–H, N–H or F–H bond?** — a hydrogen atom covalently bonded to nitrogen, oxygen or fluorine, so the H carries a strong δ+ charge (a hydrogen-bond **donor**).
2. **Does the molecule (or its neighbour) carry a lone pair on N, O or F?** — the electron pair that is attracted to the exposed δ+ hydrogen (a hydrogen-bond **acceptor**).

- If both are present, a hydrogen bond can form — with the δ+ hydrogen sitting on the lone pair, roughly in line with the bond to the atom it came from.
- To hydrogen bond **with itself**, a molecule needs **both** features: a donor O–H/N–H/F–H **and** a lone pair on N, O or F.

MOLECULE	KEY GROUPS	HYDROGEN BONDS WITH ITSELF?
ethanol, CH_3CH_2OH	O–H; 2 lone pairs on O	Yes — O–H···O bonds between molecules
methoxymethane, CH_3OCH_3	C–O–C is polar; lone pairs on O — but no O–H	No — only permanent dipole–dipole and London forces between its own molecules
ethanoic acid, CH_3COOH	O–H; C=O with lone pairs	Yes — forms hydrogen-bonded **dimers**
methylamine, CH_3NH_2	N–H; lone pair on N	Yes — same pattern as ammonia
propanone, CH_3COCH_3	C=O with lone pairs — but no O–H	No — though it can accept hydrogen bonds **from water**
chloroethane, CH_3CH_2Cl	Polar C–Cl, but H is not bonded to N, O or F	No — chlorine lone pairs are too diffuse to hydrogen bond

KEY POINT

Ethanol vs methoxymethane — same formula, 100 °C apart

Both are C_2H_6O, so both have the same electrons and almost identical London forces. Ethanol boils at **78.5 °C**, methoxymethane at **−24.8 °C**. Ethanol contains an **O–H group**, so its molecules hydrogen bond to each other; methoxymethane's oxygen is buried between two carbons with **no O–H bond**, so only dipole–dipole and London forces hold its molecules together. Over 100 °C of boiling temperature — all from one hydrogen bond.

Dimers: hydrogen bonding between carboxylic acids

- Molecules of **ethanoic acid dimerise** through hydrogen bonding when dissolved in certain organic (non-aqueous) solvents.
- The O–H hydrogen of one molecule sits on a lone pair of the **C=O oxygen** of the other — and vice versa — so **two** hydrogen bonds lock the pair into an eight-membered ring.
- Measuring a relative molecular mass of about **120** instead of 60 in such a solvent is evidence for the dimer $(CH_3COOH)_2$.

$$2CH_3COOH \rightleftharpoons (CH_3COOH)_2$$

Hydrogen-bonded dimer of ethanoic acid, formed in non-aqueous solvents

WATCH OUT

Predicting — the examiner's traps

A polar molecule is **not** automatically hydrogen-bonded: the H must be bonded **directly to N, O or F**. HCl is very polar but cannot hydrogen bond with itself. Equally, a molecule with no O–H/N–H/F–H of its own can still accept hydrogen bonds **from water** — that is why methoxymethane and propanone are water-soluble even though they never hydrogen bond with themselves.

SPEC 7.5 ≈ 45 MIN

Intermolecular Forces and Physical Properties

THE SPECIFICATION SAYS

understand, in terms of intermolecular forces, physical properties shown by substances, including: i the trends in boiling temperatures of alkanes with increasing chain length ii the effect of branching in the carbon chain on the boiling temperatures of alkanes iii the relatively low volatility (higher boiling temperatures) of alcohols compared to alkanes with a similar number of electrons iv the trends in boiling temperatures of the hydrogen halides HF to HI

Boiling trends of alkanes, alcohols, noble gases and the hydrogen halides — all in terms of forces.

From 7.1: intermolecular forces are far too weak to influence **chemical** properties — but they govern the **physical** ones. Every explanation below has the same skeleton: *identify the forces present, compare their strengths, and match the result to the property.*

i — Boiling temperatures of unbranched alkanes

- The most significant intermolecular attraction between alkane molecules is the **London force**.
- The boiling point of alkanes **increases as molecular mass increases**, because:

1. when molecular mass increases, the **number of electrons per molecule increases** — hence the instantaneous and induced dipoles also increase;
2. when the chain length increases, the **number of points of contact** between adjacent molecules increases. The more points of contact there are, the greater the overall London force of attraction.

ii — The effect of branching

- **Branched-chain alkanes have lower boiling temperatures than their unbranched isomers.**
- Branching lowers the melting and boiling temperature because **side chains interfere with the packing together of molecules** and reduce the formation of London forces.

NAME OF ALKANE	CONDENSED STRUCTURAL FORMULA	BOILING TEMPERATURE / K
pentane	$CH_3CH_2CH_2CH_2CH_3$	309
2-methylbutane	$CH_3CH(CH_3)CH_2CH_3$	301
2,2-dimethylpropane	$C(CH_3)_4$	283

Same molecular formula (C_5H_{12}), same electrons — the boiling temperature falls as the chain becomes more compact and the molecules make **fewer contact points**.

iii — Alcohols compared with alkanes

- Alcohols contain an **–O–H group** and can therefore form intermolecular **hydrogen bonds** in addition to London forces.
- Alcohols have a **higher boiling temperature (lower volatility)** than alkanes with the same number of carbons — or a similar number of electrons — because in alkanes only weak London forces are present, whereas alcohols hydrogen bond between their molecules.

ALKANE	BOILING POINT / °C	ALCOHOL	BOILING POINT / °C
methane	−164	methanol	65
ethane	−89	ethanol	79
propane	−42	propan-1-ol	97
butane	−0.5	butan-1-ol	117
pentane	36	pentan-1-ol	138
hexane	69	hexan-1-ol	156

ALCOHOL	ENTHALPY CHANGE OF VAPORISATION / KJ MOL^{-1}
CH_3CH_2OH	38.6
$CH_3CH_2CH_2OH$	47.5
$CH_3CH_2CH_2CH_2OH$	52.4
$CH_3CH_2CH_2CH_2CH_2OH$	57.0
$CH_3CH_2CH_2CH_2CH_2CH_2OH$	61.6

ALKANE	ENTHALPY CHANGE OF VAPORISATION / KJ MOL^{-1}
$CH_3CH_2CH_3$	15.7
$CH_3CH_2CH_2CH_3$	21.0
$CH_3CH_2CH_2CH_2CH_3$	26.4
$CH_3CH_2CH_2CH_2CH_2CH_3$	31.6
$CH_3CH_2CH_2CH_2CH_2CH_2CH_3$	36.6

- The enthalpy change of vaporisation is a measure of the amount of energy required to **completely separate the molecules of a liquid into a gas** at the same temperature — the greater the enthalpy change of vaporisation, the greater the force of attraction between the molecules.

- These tables show that the predominant intermolecular force in the first few members of the alcohols is **hydrogen bonding**. However, **London forces predominate as the chain length increases** — the growing non-polar chain contributes more and more.

DEFINITION

Volatility

A measure of **how easily molecules escape from liquids**. Volatile substances have a **low boiling temperature**.

- The **viscosity** of alcohols also increases as the number of hydrogen bonds per molecule increases: propane-1,2-diol (two –OH groups) is more viscous than propan-1-ol (one), and propane-1,2,3-triol (glycerol, three –OH groups) is more viscous still.

$$CH_3CH_2CH_2OH \rightarrow CH_3CH(OH)CH_2OH \rightarrow CH_2(OH)CH(OH)CH_2OH$$

propan-1-ol → propane-1,2-diol → propane-1,2,3-triol — increasing viscosity as the hydrogen bonding increases

The noble gases

- Noble gases are **monatomic** — single atoms, with only weak van der Waals (London) forces between them.
- From helium to radon, the size of the electron cloud increases. This increases the strength of the London forces between the atoms.
- Therefore the boiling temperature **increases down the group**, as more energy is required to overcome the forces (see the table in 7.1: 4.3 K for helium → 211 K for radon).

iv — Boiling temperatures of the hydrogen halides HF to HI

HYDROGEN HALIDE	HF	HCL	HBR	HI
Boiling temperature / K	293	188	206	238

- **HF is the anomaly.** Its boiling temperature is far above the trend because its molecules are hydrogen bonded — the strongest intermolecular force in the set.
- From HF to HCl there is a **large drop**: hydrogen bonding disappears, leaving only permanent dipole–dipole and London forces.
- After the drop to HCl there is a **steady upward trend from HCl to HI**. This is because, despite the **decrease in permanent dipole–dipole forces** between the molecules, there is an **increase in the instantaneous dipole–induced dipole (London) forces** due to the increase in the size of the atoms (Br and I hold many more electrons than Cl).
- Once again the London forces win out over the permanent dipole — the same conclusion reached in 7.1.

EXAM ALERT

The four-mark boiling-trend answer

A complete explanation always names the forces and both factors: e.g. *HCl has permanent dipole–dipole forces and London forces; HI boils higher than HCl because I has many more electrons, so the instantaneous dipole–induced dipole forces are stronger — this outweighs the weaker permanent dipole in HI.* Naming only "stronger intermolecular forces" scores nothing.

SPEC 7.6 ≈ 35 MIN

Solvents and Solubility

THE SPECIFICATION SAYS

understand factors that influence the choice of solvents, including: i water, to dissolve some ionic compounds, in terms of the hydration of the ions ii water, to dissolve simple alcohols, in terms of hydrogen bonding iii water, as a poor solvent for compounds (to include polar molecules such as halogenoalkane), in terms of inability to form hydrogen bonds iv non-aqueous solvents, for compounds that have similar intermolecular forces to those in the solvent

Hydration of ions and hydrogen-bonding alcohols — and why oil floats on water.

What has to happen for a substance to dissolve

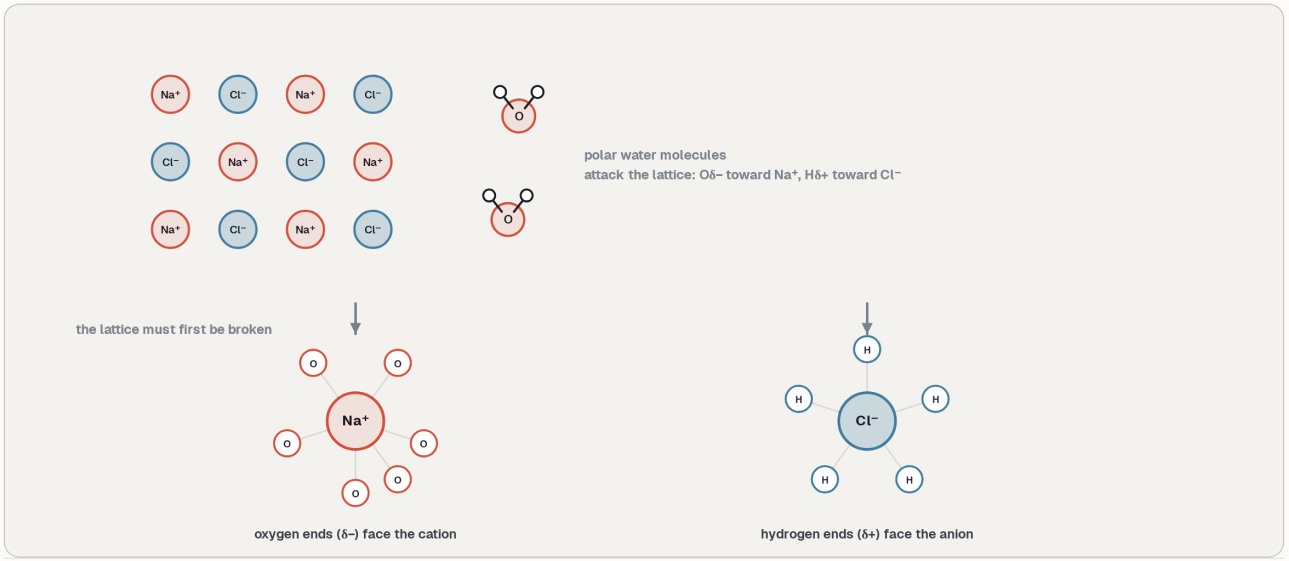
1. The **solute particles must be separated from each other**, and then become surrounded by **solvent particles**.
2. The force of attraction between solute and solvent particles must be **strong enough to overcome** the solvent–solvent forces and the solute–solute forces.

- So dissolving is a contest of attractions: the new solute–solvent attractions must pay for breaking the old ones.

i — Water dissolving ionic compounds: hydration of the ions

- In an ionic compound, the positive and negative ions are held in a **lattice structure** by strong electrostatic forces of attraction.
- Energy is required to break down the lattice — this is the **lattice energy**.
- **Water molecules are polar**: the hydrogen atoms carry a δ+ charge and the oxygen atoms carry a δ− charge.
- Attractive forces between the water molecules and the ions in the lattice **break up the lattice** to form individual ions, which spread throughout the water.
- The water molecules arrange themselves so that **positive ions are close to the negative oxygen** ends of water, and **negative ions are close to the positive hydrogen** ends.

SPEC 7.6 · HYDRATION OF THE IONS **How water dissolves an ionic solid** NaCl in water



MECHANISM Ion–dipole attractions release the HYDRATION ENTHALPY as each ion becomes surrounded by oriented water molecules. The compound dissolves when hydration enthalpy ≥ lattice energy — otherwise the energy released is insufficient to break up the lattice.

FIG — *Hydration of the ions: each Na^+ is surrounded by water with its δ− oxygen facing in; each Cl^- with its δ+ hydrogens facing in. The ion–dipole attractions released as **hydration enthalpy** pay for breaking the lattice.*

- This process is known as **hydration**, and the energy released during it is the **hydration enthalpy**.
- If the hydration enthalpy is **more than or equal to** the lattice enthalpy, the compound **will dissolve** in water. If the hydration enthalpy is **less than** the lattice enthalpy, the compound will **not dissolve** — the energy released is insufficient to break up the lattice.

KEY POINT

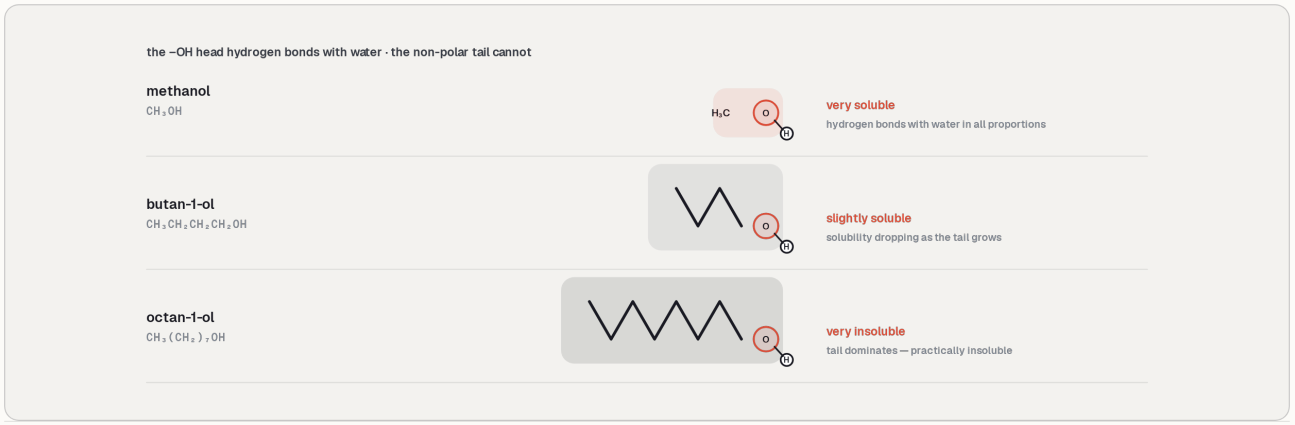
The solubility balance

dissolves when **hydration enthalpy ≥ lattice enthalpy**. The same bookkeeping explains why some ionic compounds are insoluble: their lattices are too strong (or their ions too poorly hydrated) for the water to pay the bill.

ii — Water dissolving simple alcohols: hydrogen bonding

- Alcohols with relatively **short carbon chains dissolve in water**, because the alcohol can form **hydrogen bonds with water** — stronger than the London forces between the carbon chains of the alcohol molecules.
- The solubility **gradually decreases with increasing carbon chain length**: the **non-polar hydrocarbon part becomes bigger**, which increases the strength of the London forces between the carbon chains of the alcohol molecules.
- More than one **–OH group** in the molecule increases solubility, since **more hydrogen bonds** can be formed between the alcohol and water — compare propane-1,2,3-triol (glycerol), which mixes with water in all proportions, with octan-1-ol, which barely dissolves.
- **Carboxylic acids dissolve in a similar way to alcohols** — the –COOH group hydrogen bonds with water just as an –OH group does.

SPEC 7.6 · ALCOHOLS IN WATER **Solubility falls as the non-polar tail grows** — like dissolves like



TREND Water–alcohol hydrogen bonds must replace water–water hydrogen bonds AND alcohol–alcohol London forces. Short chains: hydrogen bonding wins. Long chains: the London forces between tails dominate and the alcohol separates out. Carboxylic acids behave the same way.

FIG — *The hydrophilic –OH head hydrogen bonds with water; the growing non-polar tail can only offer weak London forces. Methanol mixes with water in all proportions; octan-1-ol is essentially insoluble.*

iii — Water as a poor solvent: compounds that cannot hydrogen bond with it

- Non-polar molecules **do not dissolve in water**: the forces between non-polar molecules are much weaker than the hydrogen bonds in water, and the attraction between alkane and water molecules is **not strong enough to disrupt the hydrogen bonding** between water molecules.
- So the two substances remain as **separate layers** — oil floats on water.
- Many **polar compounds also have limited solubility** in water. Compounds like **halogenoalkanes** cannot disrupt the water structure even though the carbon–halogen bond is polar — because the **bulk of the molecule (the hydrocarbon chain) is not polar**.
- **Ethoxyethane ($CH_3CH_2OCH_2CH_3$)** is polar, yet it is almost totally immiscible with water: the force of attraction between ethoxyethane and water molecules is not large enough to replace the relatively strong hydrogen bonding between water molecules.

WATCH OUT

Polar ≠ water-soluble

The specification deliberately includes polar molecules such as halogenoalkanes under "poor solvents". A single polar bond in a largely non-polar molecule is not enough — the molecule as a whole must offer attractions **at least as strong** as the water–water hydrogen bonds it must replace.

iv — Non-aqueous solvents

- Two **non-polar liquids will mix completely** — for example hexane and octane.
- Both liquids contain weak London forces, and these extend throughout the whole structure when they are mixed: there is no strong solvent–solvent force for the solute to fight.

Like dissolves like

- **Non-polar solutes** dissolve in **non-polar solvents**.

- **Polar solutes** dissolve in **polar solvents** such as water (H_2O) or ethanol (C_2H_5OH).
- **Ionic solutes** such as sodium chloride generally dissolve in **polar solvents** — but not in non-polar solvents, which cannot hydrate the ions.

TIP

Choosing a solvent in practice

Match the dominant force: ionic and hydrogen-bonding solutes → water; oily, non-polar solutes → hexane and other non-aqueous solvents; everything in between → ethanol, which offers both an –OH and a hydrocarbon chain. This is exactly how recrystallisation solvents are chosen in the laboratory.

END OF TOPIC 07

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02 · Atomic Structure and the Periodic Table
03 · Bonding and Structure
04 · Introductory Organic Chemistry and Alkanes
05 · Alkenes
06 · Energetics
08 · Redox Chemistry and Groups 1, 2 and 7
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